Jyotsna M | Sr. Data Engineer

jyotsna.apple9@gmail.com | 551-247-9790 [Tableau Portfolio] [GitHub - Capstone Project]

Professional Summary :

Over 12+ years of experience in Data Engineering and Data Science, specializing in processing and analyzing large volumes of structured, semi-structured, and unstructured data, with strong hands-on expertise in Apache Spark, Scala, Spark SQL, and data validation.

Technical Skills:

ETL: Azure Data Factory, Azure Databricks (PySpark), AWS Glue
Orchestration: Apache Airflow
Reporting: Power BI, Tableau, Looker, SQL reporting
Languages: SQL, Python, PySpark, Spark SQL
Cloud Data Warehouse: Azure Synapse Analytics, Amazon Redshift, Google BigQuery, Snowflake
Database: SQL Server, Azure SQL Database, Amazon RDS, PostgreSQL, MySQL

EDUCATION:

Masters in IT (Data Science) Saint Peter's University –(**3.9** GPA) New Jersey

CERTIFICATIONS:

AWS, Snowflake SnowPro, PSM-1(Professional ScrumMaster), Azure, AINS – 101, GCP-Google Cloud Platform ,PSPO-1(Professional Scrum Product Owner) **DataIku ,DBT ,**

- Databricks Certified DataEnginer Associate Certification
- Databricks Certified DataEngineer Professional Certification
- Datbricks Certified Data Analyst Associate

EXPERIENCE:

W.R.Berkley Corporation(P &C Insurance) Property & Casuality - Richmond(Virginia)

Data Engineer : **May 2023 - Present**

- Involved in building Data Lakes, validating, transforming, and publishing data to support Policy & Claims management workflows for property and casualty insurance operations.
- Recreated existing application logic and functionality in the Azure Data Lake, Data Factory, SQL Database, and SQL Data Warehouse environment, integrating policy and claims data from Guidewire Policy Center and Guidewire Claims Center into centralized storage.
- Designed and developed ETL pipelines in the Azure cloud to retrieve customer data from APIs, process it into RDBMS SQL Server and Azure SQL Database, including claims data from Guidewire systems.

- Designed and developed ETL processes to integrate data from multiple sources into a centralized data warehouse, supporting policy and claims management in the insurance domain, ensuring seamless data flows from Guidewire Policy Center and Guidewire Claims Center.
- Implemented data visualization tools like Tableau and Power BI to create interactive dashboards for real-time monitoring of key performance indicators (KPIs), including claims processing and policy metrics drawn from Guidewire.
- Developed Extract, Transform, Load (ETL) and Extract, Load, Transform (ELT) processes to extract data from various sources, perform transformations, and load data into target systems such as RDBMS SQL Server and Azure SQL Database, handling large-scale policy and claims data processing.
- Designed and implemented migration strategies for traditional systems on Azure (Lift and Shift/Azure Migrate, other third-party tools), migrating policy and claims data from Guidewire systems and SQL Server into Azure environments, using Azure Data Lake, Data Factory, and Azure SQL Data Warehouse.
- Designed and implemented end-to-end data solutions (storage, integration, processing, visualization) in Azure, incorporating policy and claims data into Azure Data Lake and leveraging PySpark to handle large-scale data processing. Utilized GitHub for version control and collaborative development, managing source code and maintaining a streamlined CI/CD pipeline for ETL process automation.
- Implemented Apache Airflow to orchestrate and automate complex ETL workflows, scheduling and monitoring data pipelines to ensure reliable and timely delivery of policy and claims data from multiple sources to the centralized data lake and warehouse.
- Developed and managed databases, wrote SQL queries, and performed ETL tasks related to policy and claims data in *RDBMS SQL Server*, implementing metrics-based scoring models to provide insights for data engineering projects.
- Worked on *Data Mapping* for multiple organizational units within the database, integrating policy and claims data from *Guidewire* into various workspaces for analysis and reporting.
- Created *Data Dictionaries*, performed data validation, and prepared data for publishing Power BI and Tableau reports in the cloud, handling large datasets from *Guidewire* and using Azure Data Lake for claims and policy data. Built ETL solutions using *Databricks*, executing code in Notebooks against policy and claims data in *Azure Data Lake* and *Delta Lake*, and loading the processed data into *Azure Data Warehouse* for further analysis.
- Used *PySpark* and *Spark SQL* for data extraction, transformation, and aggregation of policy and claims data from multiple file formats, supporting large-scale analysis of insurance operations.
- Worked on Data Modeling to redefine entities,relationships and attributes as per specifications after analyzing database systems to make sure it aligns with business needs
- Developed Pyspark Scripts from Source System to ingest data in reload, append, and merge mode into Delta tables in Databricks
- Implemented robust data ingestion pipelines to seamlessly transfer data from various sources into **ADLS Gen2** using Azure Data Factory. This ensured efficient handling of large volumes of structured and unstructured data
- Designed and implemented end-to-end data pipelines using Spark and Scala to automate data ingestion, transformation, and loading processes.
- Monitored and managed data pipelines using Azure Monitor..
- Used CI/CD tools likle Jenkins,Gitlab and Azure Devops for automating tasks such as building,testing and deploying applications helping teams streamline development workflows and improve efficiency

Skills: Data Warehousing, PySpark, Azure Databricks · Azure Data Factory(ADF), Azure Data Lake, Azure Key Vault, SQL Azure Management Studio, HiveMetastore, Tableau, Unity Catalog, Power BI, Agile Methodology, Data Wrangling, Data Mining, Python, Azure Synapse,Data Modeling, data governance, Azure Data Bricks, Azure SQL Database, Guideware DataMart,Spark,Scala, Microsoft Office Suite (Word, Excel, and PowerPoint), Jupyter Notebook,JIRA,Scrum, **,**Delta Lake,HIPAA,PII, Policy & Claims,Guidewire Policy Center, Guidewire Claims Center,RDBMS SQL Server,Snowflake,Github

Bayer (Saint Louis-Mo) - Pharma and Health care
Data Engineer : **June 2022 – May 2023**

- Worked as an IT Technical Liaison, gathering requirements from business partners and leveraging relationships with IT and business teams to enhance and operate systems effectively, incorporating GCP services for optimal performance.
- Implemented advanced web and application tracking strategies.Developed pipelines using Cloud Build and GitOps practices (e.g., using ArgoCD) to automatically deploy updates to insurance microservices and data-processing jobs on GKE.
- Developed data pipelines for transferring data from Intelex to the CSW platform, leveraging Apache Beam and Google Cloud Dataflow for ETL processes. Ensured efficient data integration and seamless data flow between systems.
- Developed custom event tracking and conversion optimization solutions with Google Analytics
- Successfully migrated data from on-premise SQL Server to cloud databases. Utilized Google Cloud SQL for scalable and reliable database management, improving data accessibility and performance.
- Validated and tested data records before production migration. Ensured data quality and consistency in BigQuery and across GCP environments to maintain data integrity.
- Utilized Snowflake for source-to-target data mapping, enhancing data warehousing capabilities. Complemented this with Google Cloud SQL for improved data management and analysis.
- Collected and managed metadata using BigQuery for data analytics. Employed Looker for visualization, providing comprehensive insights and enhancing reporting capabilities.
- Verified data post-migration to ensure its integrity and correctness in GCP environments. Focused on robust data pipeline management and validation to ensure data accuracy.
- Designed and implemented ETL pipelines using Google Cloud Dataflow and Airflow/Composer for Automated data processing tasks to increase efficiency and streamline workflows.
- Facilitated data integration and replication across platforms using GCP infrastructure. Leveraged Pub/Sub (or Kafka) to ensure seamless data flow and management.
- Developed infrastructure automation scripts with Terraform and Python. Optimized deployment processes for GCP services, enhancing operational efficiency.
- Employed scripting languages like Python and Bash for automation and development. Integrated these scripts with GCP tools to improve functionality and automation.
- Actively participated in scrum meetings, contributing to project planning and delivery. Focused on integrating and optimizing data workflows in GCP for effective project execution.

Environment : Jira, GitHub, SQL, Google BigQuery, Google Cloud SQL, Agile, Python, DBT, Google Cloud Dataflow, Pub/Sub, Google Compute Engine, Terraform, Snowflake, Matillion, Perl, Powershell, Dataflow, Google Cloud Functions, Google Cloud Pub/Sub, Spark SQL, Scala, Power BI, GKE, Big Query, Cloud SQL, IAM, Looker, GCS, Firebase, Apache Beam, apache spark,BigQuery,dataflow, Airflow/Composer,Hadoop, **Kubernetes(GKE), Google Analytics**

Verizon (Texas) **Feb 2022 – June 2022**
GCP Data Engineer /Big Data Engineer

- Collaborate with DevOps team members to ensure successful production deployments, as well as, ensure security and effective monitoring in production environments
- Created Oozie workflows for Hadoop-based jobs including hive. Perform Git code check-in, Git code review
- Created Hive external tables and loaded data into tables and query data using HQL
- Code changes in the Hql scripts and bteq scripts. Pushing those changes and deploying the code into Production.
- Read the data files from the source (GCP) and do the code changes in Job Modules and Hql Scripts and its dependent tables as per the Business user's need.
- Managed data ingestion processes, including data loading, validation, and integration from various sources into Snowflake, BigQuery.Implemented data governance policies and access controls to ensure data security and compliance with regulatory requirements.Experience in GCP Dataproc, GCS, Cloud Functions, Big Query
- Used Git for version control, JIRA for task tracking, and actively involved in Agile Methodology
- Experience with DBT for running scheduled ELT workflows. Involved in checking for any Job failures in OOZIE.Fixing and debugging the failures, finding the root cause, and helping smoother workflow of jobs.
- Good experience with Teradata SQL querying and checking any issues on the production side by running queries in Teradata and Debugging through code to fix any bugs
- Used Information security to protect sensitive information from unauthorized access and enforcing security polivcies and ensuring compliance with regulatory requirements(ex:HIPAA,PCI,GDPR)
- Used Git for version control, JIRA for task tracking, and actively involved in Agile Development Methodology
- Used IntelliJ for making code changes to bteq and Hql scripts and its dependent tables. Commit changes and push into the Master branch. Create an Artifactory and raise a Jira request for deployment. Collaborate with the Production support team to stop the job workflow during deployment and rerun the jobs afterward. Give them the document to run the Scripts and follow the steps for deployment
- Migrated SQL server database to Big Query and used Power BI for reporting

Environment: Jira, Git Hub, IntelliJ, Hadoop, Hive, Oozie, Teradata, Hql, Bteq, Gitbash, GitHub, GCP, ServiceNow, DBT, Terraform, Snowflake, Google Big Query, Agile methodology, Cloud Functions,GCP Cloud storage,Sql Server,CI/CD,DataProtection,GCP DataProc,GCS,Scrum,Agile,DBT,Github,Airflow

State Of California (Accenture) - CA **June 2021 – Feb 2022**
Data Engineer

- Worked closely in Data Science and Data Engineering work Streams and Comfortable and experienced working in Agile Software Development Life Cycle (SDLC) environments.
- Done AWS Deployment using cloud formation and Terraform.Migrating data from different record-keeping systems into AWS and Collaborating with system architects, design analysts, and others to understand business requirements.

- Successfully executed **end-to-end migration from SQL to PostgreSQL**, using **AWS DMS** for data replication (full load + CDC) and **AWS Schema Conversion Tool (SCT)** for schema and stored procedure transformation.
- Migrated datasets from on-premises and legacy systems to **AWS Aurora PostgreSQL**, resolving schema conflicts, optimizing data types, and ensuring referential integrity.
- Leveraged **Athena** in conjunction with AWS QuickSight for interactive data visualization and business intelligence reporting.
- Integrated Databricks with various data sources such as AWS S3, and on-premises databases, ensuring seamless data flow and enhancing collaborative efforts within Databricks.
- Implemented data storage and processing solutions on AWS using services like AWS S3, Amazon DynamoDB, and Amazon Redshift.Glue for data integration, ETL (Extract, Transform, Load), and data cataloging tasks.
- Proficient in using AWS CloudWatch to monitor resource utilization, application performance, and overall system health.
- Used Informatica Power Center for (ETL) extraction,transformation and loading data from heterogenous source system into target database
- Designed and implemented scalable data warehouse solutions using AWS Redshift.
- Developed and maintained ETL pipelines using AWS Glue to integrate data from diverse sources into the data warehouse.
- Configured **DynamoDB** tables with appropriate partition keys and sort keys to optimize data retrieval and storage.
- Automated data processing tasks using AWS Lambda, enhancing ETL workflows.Ensured data security and compliance by implementing AWS IAM roles for access control.
- Created data models and optimized data structures for reporting and analytics in collaboration with data scientists and analysts.
- Utilized **GitLab** CI/CD to automate the build, test, and deployment processes, reducing manual intervention and errors
- Implemented cross-region replication and automated backups for high availability and disaster recovery.
- Worked on EDD Fraud Data Analysis. Work with the Application team in understanding the requirements and converting them into technical details.
- Integrated **API Gateway** with AWS Lambda, enabling serverless backend processing for APIs and reducing infrastructure management overhead
- Implemented machine learning algorithms, including decision trees, random forests, and gradient boosting in R for predictive analytics, and utilized **AWS SageMaker** for training and deploying these models
- Developed state machines using **AWS Step Functions** to manage the execution flow of serverless applications, ensuring reliable and scalable operations
- Leveraged **AWS SageMaker** for model training, tuning, and deployment, integrating machine learning workflows into the broader data processing and analytics pipeline.
- Developed and maintained test suites using **pytest** to ensure the robustness and reliability of Python applications
- Configured **Lake Formation** to integrate with AWS IAM and AWS KMS, ensuring secure access and management of encryption keys.
- Designed and developed ETL pipelines using Spark and Python, and serverless ETL pipelines using AWS Glue, EC2, S3, Redshift, Lambda, and Triggers to perform data cleaning, and data profiling efficiently

Environment: AWSQuicksight, Redshift, Athena, Tableau, S3, python, PySpark, Glue, SFTP, Git, WinSCP, Putty,DMS, CloudWatch trail, RDS, IAM, SQS, Lambda, Cloud Watch Log, DataBricks, EMR, ServiceNow, SQL Server Management Studio, Python, Spark SQL, Scala, Terraform,AWS Aurora,Informtic Power Center ,BOTO3, SNS , Datawarehouse, AWS SAGEMAKER , JIRA,Scrum,Agile,DBT,Kinesis,DynamoDB.

Kmart Retail (New Jersey) **Jan2021 – June 2021**
GCP Data Engineer/Data Scientist

- Worked as a Data Scientist and Machine Learning Engineer on projects that involved the utilization of machine learning and deep learning algorithms for the analysis of customer data to improve company profit. Applied NLP techniques for product recommendation. **Sentiment Analysis using NLP (Python, Tableau):** The objective was to extract the aspects and opinion indicators using Text blob, NLTK, and Spacy libraries. Post-data preprocessing, the polarity and subjectivity of the sentences were determined using a text blob.
- Extraction of N-Grams (Bi-Grams and Tri-Grams) using Spacy and their corresponding sentiments.
- **Used Python libraries like Scikit-Learn** for building and evaluating models such as linear regression, decision trees, and clustering algorithms, **TensorFlow and Keras f**or developing deep learning models, including neural networks for complex tasks like image and text classification

Tools: Python: NumPy, Pandas, NLTK, And Tableau, AWS, R, Google Cloud, Anaconda, AWS Sage maker,Jupyter Notebook,Python

Keurig, Burlington MA **Aug 2017 – Dec 2020**
Data Engineer/AWS Snowflake Data Engineer.

- Experience with **Snowflake Multi-Cluster Warehouses.** Involved in Migrating Objects from Oracle to Snowflake. Created a Snow pipe for continuous data load.
- Used Temporary and Transient tables on different datasets. Worked with both Maximized and Auto-scale functionality. Created a Snow pipe for continuous data load.
- Used COPY to bulk load the data. Involved in different data migration activities. Created internal and external stages and transformed data during load.
- Used the FLATTEN table function to produce a lateral view of the VARIANT, OBECT, and ARRAY columns. Worked with both Maximized and Auto-scale functionality.
- Hands-on experience in setting up workflow using Apache Airflow and Oozie Workflow engine for managing and scheduling Hadoop jobs.
- Good understanding and related experience with Hadoop stack internals, Hive, Pig, and Map/Reduce.
- Designed various Jenkins jobs to continuously integrate the processes and executed the CI/CD pipeline using Jenkins. Design and implement disaster recovery for the PostgreSQL Database.
- Set up full CI/CD pipelines so that each commit a developer makes will go through the standard process of the software lifecycle and get tested well enough before it can make it to production.
- Managed datasets using **Panda data frames** and **MYSQL**. Queried the database queries using **Python-MySQL** connector and retrieved information using **MySQL DB**.
- Used Spark and **Spark-SQL** to read the parquet data and create the tables in the hive using the Scala API
- Experienced in working with Spark ecosystem using Spark SQL and Scala queries on different formats like Text files, and CSV files. Installed and configured Hadoop MapReduce, HDFS

developed multiple MapReduce jobs in Java for data cleaning and pre-processing. Used XML parsers to parse and fetch information from **XML** templates.

- Develop MapReduce/Spark modules for machine learning & and predictive analytics in Hadoop on AWS.
- Experience with SQL programming skills and Developed Stored Procedures, Triggers, Functions, and Packages.
- Efficient in developing Logical and Physical Data models and organizing data as per the business requirements using Sybase Power Designer, and ER Studio in both Online Transaction Processing (OLTP) and Online Analytical Processing (OLAP) applications.
- Experience in Data Integration Validation and Data Quality controls for ETL process and Data Warehousing using Informatica. Work with and extract data from various database sources like Oracle, SQL Server, and DB2.
- Involved in designing and deploying multi-tier applications using all the AWS services like (EC2, Route53, S3, RDS, Dynamo DB, SNS, SQS, and IAM) focusing on high-availability, fault tolerance, and auto-scaling in AWS Cloud Formation
- Worked in areas of **Data Sharing** in Snowflake. In-depth knowledge of. Snowflake **Database, Schema, and Table** structures. Experience in using Snowflake **Clone** and **Time Travel**
- Worked on ETL Migration services by developing and deploying AWS Lambda functions for generating a serverless data pipeline that can be written to Glue Catalog and can be queried from Athena.
- Extensively used ETL to transfer and extract data from source files (Flat files) and load the data into the target database. Used Informatica Power Center for (ETL) extraction, transformation, and loading data from heterogeneous source systems into target database

Environment: AWS cloud, XML, JSON, Scala, Gitlab, Git version control, JIRA, Python 3.8, AWS, R Studio, Linux, MySQL, Postgres SQL, Informatica PowerCenter 9.x/8.x, Oracle 12C, Airflow, Jenkins, Spring boot, Service Now, shell script, Airflow, Oracle sql, Microsoft Dataverse

HCS,(Hybrid Closys Software) Hyderabad, India **June 2013 – Aug 2017**
Data Engineer/ Analyst.

- Designed and developed Power BI reports for chatbot analysis. Prepared Data Visualization insights using Power BI for better user experience to HP printer Computer and Chat Bot
- Interacted with Chat Bot and found out the blockages of questions that Chat Bot can't give or direct appropriate links to the users.

  Environment: Power BI, Tableau, My SQL, MS Excel, Tableau, , Microsoft Office Suite (Word, Excel, and PowerPoint)